



1 Introduction

This report discusses the progress achieved in the third month of our project, following the success of the previous month. Our main emphasis this month was on **enhancing and optimizing lane detection, tracking, traffic sign recognition algorithms, parallel parking algorithms and intersection crossing algorithms, further improving accuracy and efficiency.** One of the key developments was the integration of vehicle motion algorithms in highway scenarios with additional vehicle to pass. This activity involved solving challenges such as continuous detection, speed modulation, and precise maneuvering. Additionally, we made significant advancements in the detection of curved lanes, improving the system's ability to navigate complex road geometries with greater precision. These developments contribute to a more robust and adaptable system for real- world driving conditions.

2 Planned Activities:

- Curve lane detection
- Highway lane detection
- Intersection crossing
- parallel parking

Curve lane detection

Status: completed

Implementation:

The curve lane detection algorithm employs a combination of computer vision and polynomial fitting techniques. The system first preprocesses camera input through region-of-interest (ROI) masking and perspective transformation to obtain a bird's-eye view. Edge detection using Canny thresholds identifies lane markings, followed by a sliding window search to locate lane pixels. A second order polynomial is fitted to these pixels to model curved lane boundaries mathematically. The algorithm continuously updates curvature radius calculations to determine steering angles, using the polynomial coefficients to predict the vehicle's required trajectory. Kalman filtering smooths frame-to-frame variations, ensuring stable curve following even with partial lane visibility.

Highway lane detection

Status: Completed

Implementation:

Uses deep learning for identifying parallel lane markings at high speeds. Maintains confidence scores for each lane, prioritizing solid over dashed lines. Temporal tracking handles lane merges and worn markings. Multiple hypothesis tracking considers several lane configurations simultaneously, selecting the most probable based on vehicle dynamics and sensor fusion to distinguish lanes from vehicle edges.

Intersection crossing

Status: Completed

Implementation:

The intersection crossing algorithm combines perception, prediction, and planning modules. The perception layer uses sensor fusion (camera) to detect traffic lights, stop signs, crosswalks, and other vehicles/pedestrians. A behavior prediction module forecasts the intentions of surrounding agents using trajectory prediction models, often employing inverse reinforcement learning. The planning component generates a safe crossing trajectory by checking clearance conditions: verifying traffic light status, ensuring intersection is clear, and validating right-of-way rules. The algorithm uses finite state machines to transition between states (approach, stop, wait, proceed, clear), with safety constraints preventing execution unless all conditions are satisfied. Priority queuing handles multi-agent scenarios at four-way stops.

Parallel parking

Status: Completed

Implementation:

The parallel parking algorithm executes a geometric path planning approach. Initial perception identifies suitable parking spaces by measuring gaps between vehicles using ultrasonic, verifying minimum length requirements. The algorithm calculates optimal entry and exit waypoints using Reeds-Shepp curves, accounting for the vehicle's turning radius and kinematic constraints. The maneuver typically involves three phases: an initial reverse arc positioning the rear wheel, a forward correction reducing the angle, and a final reverse alignment centering the vehicle. Real-time feedback from proximity sensors enables dynamic adjustments, halting motion if obstacles appear. The algorithm employs pure pursuit for steering commands, ensuring smooth execution while maintaining safety margins from surrounding vehicles.

3 General status of the project

Our project has made significant progress, overcoming real-time challenges and implementing key improvements to enhance overall performance. We have successfully completed the traffic sign detection system, complex lane detection for both straight and curved paths, and the algorithm for parallel parking and intersection crossing.

With these developments, our system can now efficiently handle environment changes, adjust speed, maintain lane stability, and adapt to changing lighting conditions. These advancements ensure greater adaptability to dynamic environments, making the system more reliable for real-world scenarios.

4 Difficulties faced during this month

A critical hardware failure significantly impacted our project execution. The power board malfunctioned during the testing phase, preventing us from deploying the algorithms on the actual vehicle platform. Consequently, we were limited to performing simulations and controlled test runs in a virtual environment rather than real-world implementation. While this allowed us to validate algorithmic logic and fine-tune parameters computationally, this hardware setback restricted our ability to demonstrate full system integration and real-time responsiveness that would have been achieved through on-vehicle testing.